Life cycle of a Star

http://www.telescope.org/pparc/res8.html

NEBULA. A nebula is a _____ of gas and dust in space. Nebulae are the _____ of stars.

STAR Main Sequence. A star is a _____ globe of gas producing its own heat and light by nuclear fusion.

Luminous= giving out light.

RED GIANT. This is a large bright star with a cool surface. It is formed during the _____ stages of the life of a star like the Sun, as it runs out of _____ fuel at its centre.

SUPERNOVA . This is the _____ death of a star, and often results in the star obtaining the brightness of 100 million suns for a short time.

NEUTRON STARS. These are produced when a super _____ explodes., Neutron stars are very _____. Typical stars having a mass of three times the Sun but a diameter of only 20 km. If its mass is any greater, its gravity will be so strong that it will shrink further to become a black _____

. **BLACK HOLES.** Black holes are believed to form from massive stars at the end of their lifetimes. The gravitational pull in a black hole is so great that _____ can escape from it, not even _____. Holes distort the space around them, and can often suck neighbouring matter into them including _____.

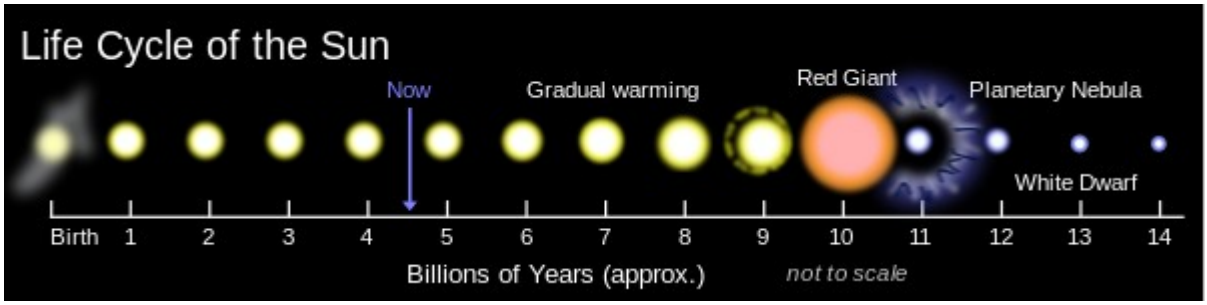
Nebula Names

Nebulae get their names for a number of reasons. Some nebulae get their name from what they look like. Some nebula are named after famous people. Some nebulae are just given numbers.

Some Famous Nebulae. Find information for these nebulae using the internet.

Name of Nebula	When was it discovered ?	How far from Earth in Light years(Ly)	Colours you can see in the picture of the nebula
Cat's eye			
Dumbbell			
Lemon slice			
Hour glass			
Keyhole nebula(Carina)			

What was used to give these nebulae their name?



Look at this diagram above. It is timeline for our sun.

How long from now will our sun last before it becomes a red giant?

How old is our sun at the moment?

Which bit of the lifecycle has not been shown here?